Applied Math 3 – Fall 2025

Assignment 4: “Game Feel” Capstone Project (30 %)

Due date: Phase 1: November 18th, 2025 at 11:59 am

Phase 2: November 29th, 2025 at 11:59 am

Phase 3: December 3rd, 2025 in class

Overview of assignment

Starting with the AM3 - A3 Flight Demo assignment hand-in as a starting point, further refine the “Game Feel” of the flight mechanics by adding effects or interaction. You should showcase the use of your math skills you have learned in this program.

Phase 1 Short proposal of your capstone design project.

Phase 2 Completion of your “Game Feel” project as defined in Phase 1.

a. **Summary Document - PDF File**
b. **Rules/Controls Page**
c. **Project Assets: Build & Project files**

Phase 3 Demonstration to the class of your AM3 project.

Areas of focus to improve “Game Feel”:

- **Input:** Refining and improving the input system through testing and iteration. Selecting values for your input responsiveness, decay and other settings that improve the control of your Pawn. You could also offer additional controller inputs, such as Xbox gamepads (it must be something I can test it with though, so nothing too esoteric).
- **Movement:** Extending the movement code to include better feeling physics, improved easing, and interactions with the environment (e.g. deflecting off the ground in a realistic way).
- **FX & Responsiveness:** Creating an improving the visuals of your Pawn to be responsive to the input. This could optionally include Particle Systems but could also just be things like wing flaps moving, engines firing/not firing, or improvements to the UI that help communicate the state and motion. This could also include integrating and controlling animations if you want.
- **Debugging and Visualization:** Including refined debugging interfaces and tools to help visualize and troubleshoot the movement.

Learning outcomes

- Refining your AM3 – A3 Flight Demo to improve the "Game Feel" through improvements to the Input, movement and FX systems.
- Demonstrate your ability to apply the math skills you have learning in this program to improve the gameplay and "Game Feel" of a project demo.
- Be able to articulate both the problems or design goals, and the solutions and how you applied them to a real-life gameplay design problem.
- Demonstrate a clear understanding of the math going from input, into movement, and finally into visual and audio refinements.

Task Breakdown

Phase 1:

Create a short proposal document that outlines what you are hoping to accomplish. It is not for marks, but you should get instructor approval before starting on the main assignment (executed in Phase 2).

Goal: Identify the project you will start from, and the goals you will complete in Phase 2.
Deliverables: A short document outlining your proposed project.

This should be 1 page (max) and include a screenshot or two and perhaps some rough sketches that help communicate your vision. Does not have to be polished but should be clear and effectively communicate your design vision, and what you are going to implement to achieve it. You should have a clear design reference (e.g. A Tie Fighter, A Pterodactyl, an Undersea Submarine, etc.)

You should clearly identify if your focus will be spread across the Input, Movement and FX evenly, or if you are focusing more on one or two of the categories. Think about how you will support your overall design goal with the features you add.

Phase 2:

Upgrade an existing project to refine it's "Game Feel" elements such as input and movement. It should have a focus on using math to support the "Game Feel".

You should be getting other students to playtest your game periodically and include notes on how you addressed the playtesting feedback in your summary document.

Starting Point:

- Suggested: Use the AM3 A3 hand-in as a starting point for this assignment.
- Alternative: Use a GS prototype or existing game demo as a starting point.

Phase 3

This will be an informal and ungraded presentation of your AM3 project to the class, followed by playtesting (if time allows).

Areas of focus to improve “Game Feel”:

There are many more, but this should give you some suggestions to get started:

- **Input:**
  - Input improvements could touch on the Input modules themselves, such as improving timing and responsiveness.
  - You could add controller or joystick support, or alternate control schemes.
  - Keep a record of your testing, the feedback you were getting, and what changes you made to respond to it.

- **Movement:**
  - Extending the movement code to include better feeling physics or tuning the physics for improved motion.
    - For example, increasing the drag based on the surface area exposed to the direction of movement to allow for swooping”.
    - Improved easing for movement and rotations to home in on a particular sense of motion.
    - Interactions with the environment (e.g. deflecting off the ground in a realistic way).
    - “Real” physics is often not the best feeling, so you might deviate from it.

- **FX & Responsiveness:** Creating an improving the visuals of your Pawn to be responsive to the input.
  - This could optionally include Particle Systems, especially if they are responding to input.
    - Engine effects scaling with throttle is a great on.
    - VFX on the wing tips can help communicate rotation and “rolls”.
    - Dust clouds being blown up from the ground when you fly too low.
    - Crashing or explosion effects. Rocket trails or smoke (if appropriate).
    - Environmental effects (floating particles in the world, dust clouds, etc.)
  - Mechanical movement of the model or “Grey box Art” model.
    - Wing or tail flaps moving as you turn.
    - Wings flap, or the head turns to follow the camera (e.g. for a bird, dragon, dinosaur, etc.).
  - Improvements to the UI that help communicate the state and motion.
    - This could also include integrating and controlling animations if you want, but 3D art assets are optional.
    - Visualizing the throttle, a compass (or 3d compass), mini map, cockpit or other UI or in game UI elements that help sell the idea of motion.

Deliverable assets will be:

1. **Summary Document - PDF File**
    - This should be a 1-page polished document that would be appropriate for a portfolio and highlight what changes you made in the project for A4 and why you made them.

    Use these section headings and address each in your summary:

    - **Game Overview & Concept**
    - **Design Improvements Made:**
        - A brief list or summary so I know what you worked on.
    - **Testing Notes:**
        - What did you learn from people playtesting your game?
    - **Rules:**
        - Why did you choose the controls you chose, how did you refine them, and how did they impact the gameplay or motion?

2. **Rules/Controls Page**
    - Include the controls as a separate page to help with playtesting. This should be polished and ready to display but can be otherwise quite simple.
    - If you did used the same project for both GD1 and AM3, you can submit the same Controls page, but submit it to both drop boxes individually to avoid confusion.
3. **Project Assets:**

    Physical and Digital games will have different submission requirements:

    a. **For digital projects:**
        - **Unreal Project/Unity Project - Zip file**
            - The complete project. I'm not grading based on this but would like it for reference and archival purposes.
        - **Exported playable build - Zip file**
            - The playable build. This is what will be primarily marked. Make sure the controls are

    b. **For physical projects:**
        - **The physical Board Game prototype.**
            - The completed prototype. You should be considering printing graphics and using more polished materials for this option.

Additional Notes on project naming:

Please hand in your UE Project in a zip file with the name AM3_A4_FirstInitialLastInitial.zip and the build with AM3_A4_Build_ FirstInitialLastInitial.zip. The Design Doc should be a PDF with a similar naming structure such as AM3_A4_Summary_DF

E.g. AM3_A4_DF.zip and AM3_A4_Build_DF.zip

Grading Summary:

Category	What is Evaluated
Input – Responsiveness & Controls	How responsive the controls are, supported input schemes, and improvements based on playtesting.
Movement – Physics & Motion	Smoothness, satisfaction, and realism of movement; tuned physics and interactions with the environment.
FX & Visual Responsiveness – Pawn & UI	How well visuals respond to input and movement, including particle effects, animations, or UI feedback.
Testing & Iteration Notes	Evidence of playtesting, recorded observations, and iterative improvements made to the project.
Design Documentation – Summary Document	Clarity, completeness, and structure of the 1-page summary, including improvements and rationale.
Rules/Controls Page	Clarity and usability of controls or rules; support for playtesting.
Presentation & Polish – Demo Playability	Overall playability, smoothness, visual/audio polish, and demonstration of improved game feel.
Application of Math Skills	Correct and effective application of math concepts to input, movement, and FX systems; demonstrated understanding.

FAQ:

Can I use project XYZ as a starting point?

- *Probably yes, check with the instructor if you are not sure. You should have the core motion and test levels already implemented however before starting on this assignment. You won't likely have time to build something from scratch.*

Can I do both Math and GD1 assignments on the same project/demo?

- *Yes, however make sure you are doing both assignments, and make it very clear which features are for each. They are similar assignments, but GD should be focused on mechanics, challenges, and level progression, while AM is focused on game feel and motion primarily. You will still have to do both summary documents, since they focus on different things. They could share a control document though.*

Can I use this to iterate on my Game Studio Prototype?

- *Yes, most likely, however you should have a starting point already. You won't have time to develop something from scratch. Check with the instructor first.*

Can I do XYZ that isn't in the assignment document?

- *If you think it fits into the spirit of the assignment, I am open to hearing pitches but check first (and soon!) to make sure.*